

Lab Experiment 33-35

Experiment 33

Write a Python program to draw a scatter plot with empty circles taking a random distribution in X and Y and plotted against each other.

Aim

To write a Python program to generate random X and Y values and represent them using a scatter plot with empty circles.

Algorithm

1. Import matplotlib.pyplot and numpy.
2. Generate random values for X and Y.
3. Create a scatter plot using the generated values.
4. Set facecolors='none' to make the circles empty.
5. Add labels and title.
6. Display the plot.

Code

```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.rand(50)
y = np.random.rand(50)

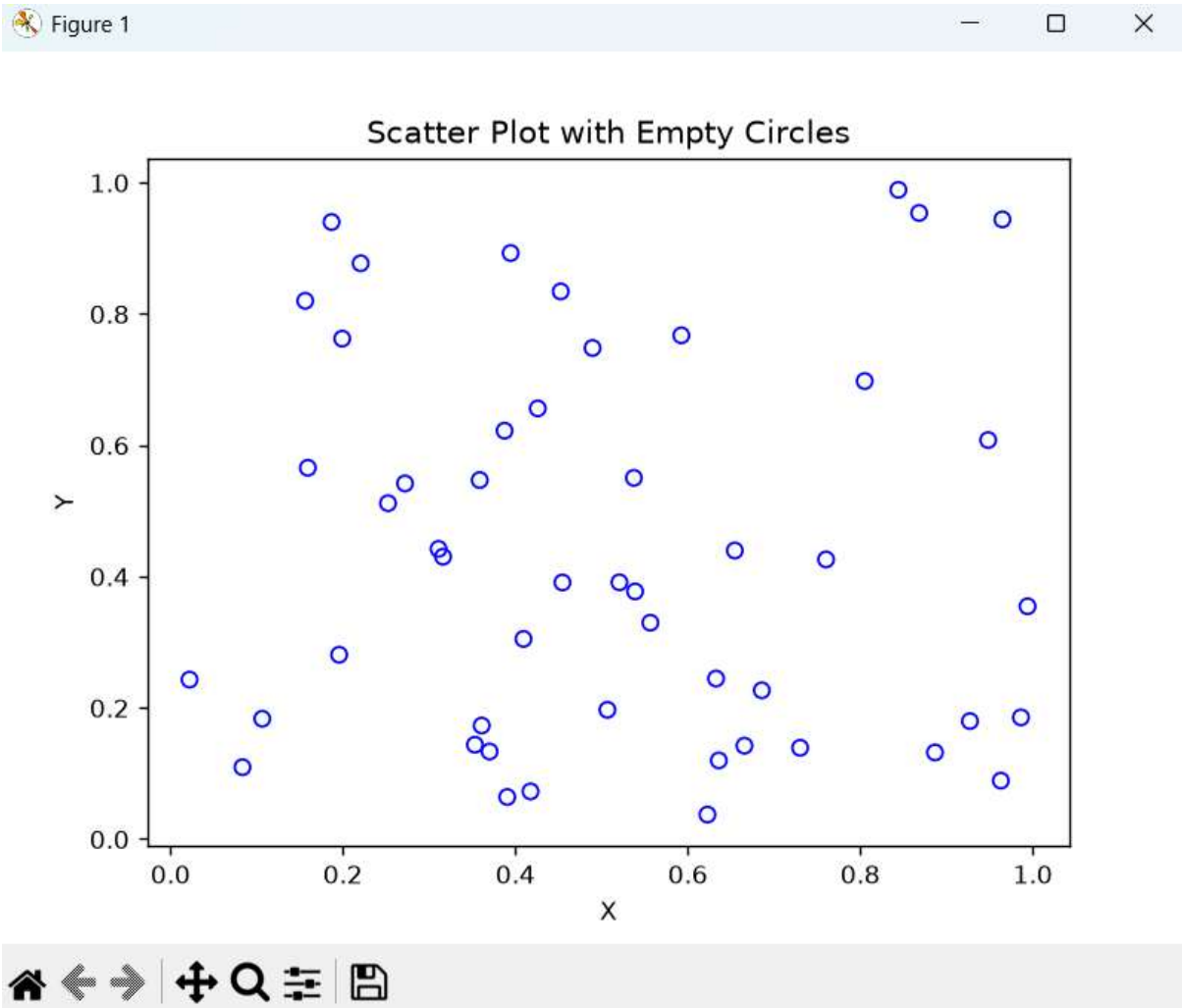
plt.scatter(x, y, facecolors='none', edgecolors='blue')

plt.xlabel("X")
plt.ylabel("Y")

plt.title("Scatter Plot with Empty Circles")

plt.show()
```

Output



A scatter plot containing randomly distributed empty circles on the X-Y plane is displayed.

Result

Thus, the Python program successfully generated a scatter plot with empty circles using random X and Y distributions.

Experiment 34

Write a Python program to draw a scatter plot using random distributions to generate balls of different sizes.

Aim

To write a Python program to generate a scatter plot with randomly distributed points having different sizes.

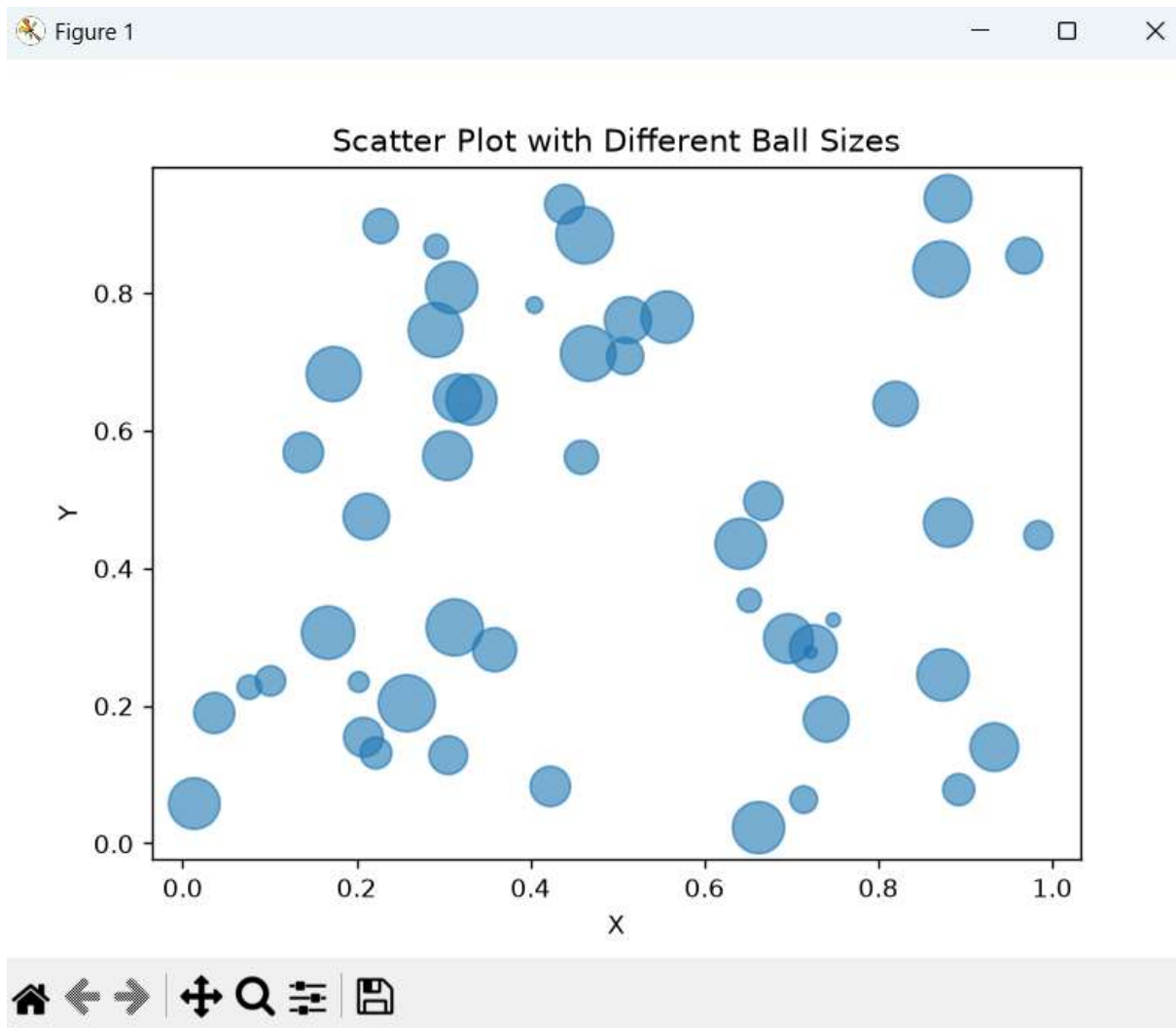
Algorithm

1. Import matplotlib.pyplot and numpy.
2. Generate random X and Y values.
3. Generate random sizes for the points.
4. Use the scatter() function to plot the points.
5. Set the point size using the s parameter.
6. Add labels and title.
7. Display the plot.

Code

```
import matplotlib.pyplot as plt
import numpy as np
x = np.random.rand(50)
y = np.random.rand(50)
sizes = np.random.randint(20, 500, 50)
plt.scatter(x, y, s=sizes, alpha=0.6)
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Scatter Plot with Different Ball Sizes")
plt.show()
```

Output



A scatter plot containing randomly distributed balls/circles of different sizes is displayed.

Result

Thus, the Python program successfully generated a scatter plot with random distributions and different-sized balls.

Experiment 35

Write a Python program to draw a scatter plot comparing two subject marks of Mathematics and Science. Use marks of 10 students.

Aim

To write a Python program to compare Mathematics and Science marks of 10 students using a scatter plot.

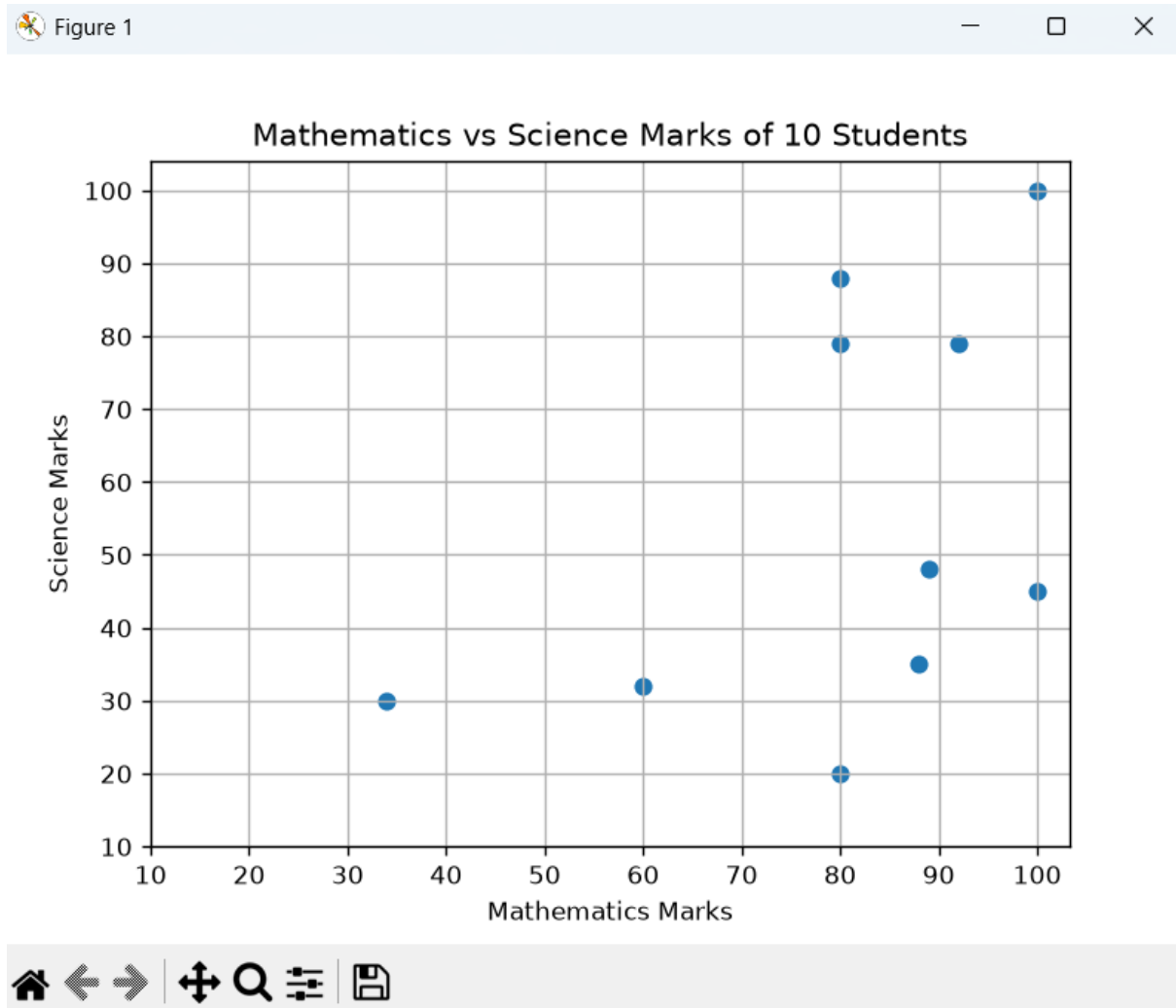
Algorithm

1. Import matplotlib.pyplot.
2. Store Mathematics marks in a list.
3. Store Science marks in a list.
4. Create a scatter plot using Mathematics marks on the X-axis and Science marks on the Y-axis.
5. Set the marks range from 10 to 100.
6. Add axis labels, title, and grid.
7. Display the scatter plot.

Code

```
import matplotlib.pyplot as plt
math_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]
science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]
marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
plt.scatter(math_marks, science_marks)
plt.xlabel("Mathematics Marks")
plt.ylabel("Science Marks")
plt.title("Mathematics vs Science Marks of 10 Students")
plt.xticks(marks_range)
plt.yticks(marks_range)
plt.grid(True)
plt.show()
```

Output



A scatter plot is displayed with Mathematics marks on the X-axis and Science marks on the Y-axis for 10 students.

Result

Thus, the Python program successfully generated a scatter plot to compare the Mathematics and Science marks of 10 students.